

Ali Haggag *Full-Stack Engineer | Real-time Systems*

✉ ali.haggag2005@gmail.com

☎ +20 102 308 1245

📍 Qena, Egypt

🌐 ali-haggag7

🔄 Ali-Haggag7

🔗 alihaggag.me

SUMMARY

Innovative **Systems & Software Engineer** specializing in architecting scalable, real-time, and offline-first systems using **Next.js, TypeScript, and NestJS 11**. Adept at designing **custom compilers/DSLs (AST parsing), 3D game physics loops**, and WebRTC/Socket.io synchronization layers. Passionate about executing comprehensive **web performance engineering (reducing TTFB/LCP)** and managing complex monorepos using **pnpm workspaces**. Equipped with a strong foundation in Agile methodologies, ready to align complex technical architectures with broader business objectives.

EDUCATION

B.Sc. in Computer Science & Artificial Intelligence

South Valley National University

Oct 2023 – Jun 2027

Qena, Egypt

- Relevant Coursework: Data Structures & Algorithms, OOP, Database Systems, Software Engineering, Operating Systems.

PROFESSIONAL EXPERIENCE

Frontend Lead

South Valley National University (Field Training)

Feb 2026 – Jun 2026

Qena, Egypt

- **Led** the frontend development of **StudentHub** 📄 (a campus platform) within an 8-person team, enforcing Agile workflows on ClickUp and strict Git PR review practices across **500+ commits** and **135+ pull requests**.
- **Architected and built 47 web screens** using React 18, Vite, and **Tailwind CSS**, implementing complete English/Arabic bilingual localization with right-to-left (RTL) layout mirroring and JWT-based role authorization.
- **Designed client-side modules and API endpoints** for the Campus Store and 6 core student services, successfully integrating them with a .NET Web API containing **110+ REST endpoints**.

Backend Intern

Web masters 📄

Jul 2025 – Aug 2025

Cairo, Egypt (Remote)

- **Engineered** scalable backend architectures using **Node.js, Express, and MongoDB**, supporting robust data flow and improving overall system maintainability for the development team.
- **Architected** secure **RESTful APIs** for a high-traffic CMS (Blog) and a URL Shortener, implementing **JWT** authentication to ensure 100% secure end-to-end data access.
- **Integrated Firebase Realtime Database** to power instant messaging capabilities, decreasing data synchronization latency by **~40%** and delivering a seamless user experience.
- **Optimized MongoDB schemas** and enforced strict data validation rules utilizing **Joi**, accelerating query execution speed by **~30%** and eliminating database inconsistencies.

PROJECTS

Logic Arena - Competitive Robot Coding Simulator 📄

Mar 2026 – Present

Next.js 16, NestJS 11, React Three Fiber, Socket.io, Redis, Prisma, Docker

- Engineered **AliScript v2.2** (a custom DSL with a Lexer → AST parser) running inside a secure server-side sandbox with strict execution limits (2,000 ops/tick).

- Architected a 20 TPS server-side 3D physics engine in a pnpm monorepo, implementing client-side state interpolation and delta compression to reduce payload size by **80%** and achieve sub-50ms latency.
- Developed a complete tournament bracket system, interactive campaign levels, match replays, and PWA capabilities.

Flurry v2.0 - Social Super App

Oct 2025 – Feb 2026

React, Socket.io, WebRTC, Google Gemini AI, MongoDB, PWA (Workbox), i18next

- Architected a real-time messaging ecosystem using WebRTC signaling over Socket.io, achieving near-zero latency (<50ms) for peer-to-peer video calls and media streaming.
- Engineered an offline-first PWA using Workbox Service Workers for local background sync, auto-syncing messages on reconnection, and integrating ImageKit CDN to reduce media payloads by **70%**.
- Integrated Google Gemini AI to generate smart summaries of long conversations and built a real-time collaborative polling system.

CS Arena - Developer Ecosystem Platform

Feb 2026 – Mar 2026

Next.js 16, TypeScript, Sanity CMS, NextAuth, Sentry, Resend, Framer Motion

- Engineered a developer discovery platform using Next.js 16 App Router, implementing a URL-first state engine and React useTransition to eliminate cascading filter race conditions.
- Architected a complete collaboration workflow with join-requests, integrating Resend for transactional emails, and tracking real-time GitHub metrics.
- Integrated Sentry for error tracking and session replays, and enforced strict rate-limiting and custom API security headers.

SKILLS

Languages: TypeScript, JavaScript (ES6+), C++, GraphQL, AliScript (Custom DSL)

Frontend & UI: Next.js (App Router), React.js, React Three Fiber (R3F), Three.js, Tailwind CSS, shadcn/ui, Magic UI, Material UI

State Management & CMS: Redux Toolkit, Zustand, React Query, Sanity CMS

Backend & Databases: NestJS 11, Node.js, Express.js, PostgreSQL, MongoDB, Prisma ORM, Supabase, Redis, Firebase

Architecture & Advanced: Compiler & DSL Design (AST), Deterministic Game Loops, Domain-Driven Design (DDD), Socket.io (Real-time), WebRTC (Streaming), Google Gemini AI, PWA (Offline-first)

DevOps, Security & Tools: pnpm Workspaces (Monorepos), Docker, Git/GitHub, Sentry, Vercel, DigitalOcean, NextAuth, Clerk, JWT, Postman, Gitlab

CERTIFICATES

Front-End Development Diploma (React.js) Sef Academy	Back-End Development Diploma (Node.js) Sef Academy
Back-End Development Training Program Web Masters	The Web Frontend Learning Guide Udemy

LANGUAGES

Arabic Native	English Professional Working Proficiency
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